Correlated-Noise Likelihoods and Next-Generation Inference for Strong Gravitational Lens Modeling*

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Technical report: correlated-noise likelihood + sampler benchmark, with the real-data verdict (P1c), A100 sampler matrix (P2c), and end-to-end recipe (P3) complete

Abstract

Drizzled space-based imaging carries strongly correlated pixel noise that the per-pixel diagonal Gaussian likelihood — the statistical core of every fast GPU lens modeling code, including GIGA-Lens (Gu et al., 2022) and its recent multi-node successor GIGA-Lens 2.0 (Huang et al., 2026) — simply ignores. On the *HST*/WFC3-IR system DESI–165.4754−06.0423 this omission produces a scale-dependent power-law slope γ (1.29 binned, 1.43 native, a 2.585 artifact on the upsampled fine product) and a two-basin γ posterior (Huang et al., 2025). We report a campaign that changes what GIGA-Lens 2.0 leaves unchanged: the likelihood and the sampler. We build a drizzle-anchored correlated-noise likelihood ($C = D^{1/2}KD^{1/2}$, convolutional whitening, operator-norm whiteness gate $e_{\mathrm{op}} \leq 0.02$) integrated with a generalized ridge marginalization of the linear source amplitudes and its Gaussian evidence (Occam) term; it reduces *exactly* to the validated diagonal stack ($|\Delta \log L| = 0$ at machine precision), matches a dense-covariance Cholesky reference to $< 3 \times 10^{-9}$ nat, and reproduces the diagonal miscalibration in miniature on drizzle mocks (median $|z(\gamma)| = 5.84$). Under a two-stage preconditioned-HMC recipe (itself a finding: it lowers $\hat{R}$ from 2.11 to 1.003), the correlated likelihood is calibrated on mocks (depth-controlled bias $|\bar{z}(\gamma)| < 0.5$ at every scale; simulation-based-calibration rank tests pass on healthy fits) and is found to be *conservative, not biased* ($\sigma_{\mathrm{fine}}/\sigma_{\mathrm{native}} = 2.45$, a characterized information cost of near-singular drizzle covariances). Independently, we deliver the first systematic sampler and multimodality benchmark on strong-lens posteriors: nine contenders across a 19-target, four-tier "posterior zoo," under a pre-registered, budget-matched, tuning-frozen protocol (370 evaluation cells). The headline is two-sided: NAUTILUS dominates sampling efficiency (2.6–307× ESS/gradient) but is disqualified on the real-lens ill-conditioned marginalization regime; no gradient method beats the GIGA-Lens baseline at budget parity; parallel-tempered HMC is the most reliable until-converged; and FLOWMC fails structurally on these geometries. On the real system, the correlated likelihood proves *necessary but not sufficient*. It flips the binned bimodality verdict: per-basin evidence swings by ∼191 nats, from favouring the steep basin

*All work reported here — code, experiments, analysis, and text — was performed by Claude Code (Anthropic), guided by Greg Benson. This document is a technical report; the three previously gated results sections (the real-data verdict §6, the A100 sampler matrix in §7, and the end-to-end recipe §8) are now filled from the completed P1c/P2c/P3 runs, with all numbers traced to the `CAMPAIGN.md` ledger.

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($\Delta \log Z = +162$ under the diagonal likelihood) to favouring the low basin ($\Delta \log Z = -29$ under the correlated likelihood), showing the upsampled steep basin to be a noise-covariance artifact (hypothesis H1 confirmed). But it *over-corrects* the slope value: the restored low-basin $\gamma = 1.103 \pm 0.008$ sits $\sim 17\sigma$ *below* the diagonal-native anchor 1.433, and the correlated slopes across scales *bracket* that anchor (1.816 fine-steep above, 1.103 binned-low below) rather than unifying onto it — a residual scale dependence pointing to further systematics in the source model and PSF (H2 rejected). On the inference axis, the A100 phase shows the two hardest real posteriors defeat off-the-shelf samplers: the 46-dimensional condition-10^{14} marginalized posterior converges under neither single-stage HMC ($\hat{R} = 3.1$) nor auto-mass MCLMC ($\hat{R} = 5.9$), only under the campaign's own two-stage re-preconditioning recipe; and on the 74-dimensional bimodal posterior every direct sampler fails mode recovery, so only per-basin SMC evidence yields a trustworthy mode weight. Every quantitative claim is traced to a JSON artifact and a gate record in `CAMPAIGN.md`.

Status of the Previously Gated Items

All three results sections that were gated on NERSC Perlmutter A100 runs are now complete: the real-data cross-scale verdict (P1c, §6), the A100 sampler matrix on the hardest targets (P2c, §7 tiers T2/T3), and the end-to-end recipe with Euclid Q1 (P3, §8). No open (gated) items remain. The pre-registered E3 kernel-variant robustness scan (Table 10) is retained as a documented follow-on (§9).

1 Introduction

Galaxy-scale strong lensing constrains the inner density profiles of massive galaxies and, through time-delay cosmography, the Hubble constant (Birrer et al., 2020; Treu, 2010). The precision of these constraints is now limited less by data volume than by modeling systematics. The Time-Delay Lens Modeling Challenge (Ding et al., 2018, 2021) established blind, truth-known benchmarking as the discipline's standard for exposing such systematics; but since its first round the community has produced few successors of comparable rigor, even as the modeling codes have accelerated by orders of magnitude. The recent code-comparison study of Galan et al. (2024) quantifies the resulting exposure: holding the data fixed and varying only the source model and modeling method shifts the recovered logarithmic density slope γ by $\sim$2% — a systematic that rivals or exceeds the statistical error bars quoted on individual systems. That same study flags two ingredients as under-addressed in fast lens modeling: correlated image noise (their motivation cites the strongly correlated noise of resampled JWST imaging) and honest posterior multimodality. Both remain open: no flagship lens-modeling paper implements a correlated-noise likelihood in a fast GPU code, and none uses flow-assisted or transport-preconditioned samplers, where nested sampling (Lange, 2023) otherwise dominates.

GIGA-Lens (Gu et al., 2022) pioneered GPU-accelerated Bayesian lens modeling with a multi-start MAP $\rightarrow$ SVI $\rightarrow$ preconditioned-HMC recipe in JAX, and has been applied to spectroscopically confirmed systems (Cikota et al., 2023; Sheu et al., 2024) and to the DESI Strong Lens Foundry (Huang et al., 2025). Its recent successor GIGA-Lens 2.0 (Huang et al., 2026) distributes exactly this recipe to 128 nodes / 512 A100 GPUs on NERSC Perlmutter, achieving — for the first time on a real system — $\hat{R} < 1.01$ on all 38 parameters of a DESI lens in $\sim$25 minutes on eight nodes. Crucially, that work *scales the computation while leaving the statistical model untouched*: the likelihood remains the per-pixel diagonal Gaussian of Gu et al. (2022), source amplitudes are solved (not marginalized) by linear inversion, and a single Gaussian SVI surrogate seeds HMC, presuming a unimodal posterior. As we detail in the campaign's positioning analysis,[1] GIGA-Lens 2.0 has zero overlap with the two axes this campaign advances: it introduces no noise covariance, no analytic marginalization or Occam factor, and no tempering, nested sampling, or normalizing flows. Faster inference of a mis-specified likelihood converges faster to the wrong posterior; our contribution is to fix what is being scaled.

The precipitating evidence is the Foundry-I reproduction of DESI–165.4754−06.0423 (Huang et al., 2025), which closed with two explicit open items. First, the drizzled *HST* products have strongly correlated pixel noise — measured lag-1 correlation $\rho(1) = 0.795$ on the $0.''04$ fine product, 0.52 at native scale, with an effective independent-pixel fraction $N_{\rm eff}/N \approx 0.017$ — that the diagonal likelihood ignores, producing a *scale-dependent* slope (1.29 binned / 1.43 native / a 2.585 artifact on the fine product) and a genuinely disconnected two-basin γ posterior on the binned product (45 of 48 chains in the low basin, 3 in the steep basin, zero inter-basin migrations). Second, HMC on such degenerate and multimodal lens posteriors is the binding computational constraint. This campaign attacks both.

We state two contributions.

1. **A correlated-noise (drizzle) likelihood with generalized ridge marginalization.** We model the pixel covariance as $C = D^{1/2}KD^{1/2}$ with D the existing heteroscedastic diagonal and K a stationary, drizzle-anchored correlation kernel fit on model-subtracted residuals; we whiten it by a truncated convolutional operator gated on its operator-norm flatness; and we integrate the whitening into an analytic ridge-marginalization of the linear (shapelet) source amplitudes that carries the Gaussian-evidence (Occam) term that the plain linear solve omits. The machinery reduces *exactly* to the validated diagonal stack in the white-noise limit, is validated against a dense-covariance Cholesky reference, is calibrated on drizzle mocks with known truth, and is tested (in §6) against a pre-registered cross-scale unification hypothesis on the real system, where it proves *necessary but not sufficient* — it removes the diagonal likelihood's upsampling artifact but over-corrects the slope value.

2. **The first systematic sampler / multimodality benchmark on strong-lens posteriors.** We assemble a 19-target "posterior zoo" spanning analytic synthetic targets, twelve Gu et al. (2022) mocks, the 46-dimensional condition-number-10^{14} marginalized real posterior, and the 74-dimensional bimodal real posterior, and benchmark nine samplers — including a normalizing-flow / neutral-transport recipe — under a budget-matched, tuning-honest, pre-registered protocol.

The correlated-likelihood construction and validation (§3, §5), the benchmark protocol and its phoenix-scale results (§4, §7), and the reproducibility substrate (§10) are the settled methodology. The three results that required NERSC Perlmutter A100 time complete the picture and are the subject of §6, the T2/T3 rows of §7, and §8. In brief: (i) on the real system the correlated likelihood is *necessary but not sufficient* — it flips the binned bimodality verdict (a ∼191-nat evidence swing, steep → low) but over-corrects the slope (binned-low $\gamma = 1.103 \pm 0.008$, $\sim 17\sigma$ below the 1.433 anchor), leaving a residual scale dependence (P1c, §6); (ii) the two hardest real posteriors need the campaign's two-stage recipe and per-basin evidence: no single-stage or auto-mass sampler converges the condition-10^{14} marginalized target, and no direct sampler recovers the bimodal mode weights (P2c, §7); and (iii) the recipe runs end-to-end and exports to a code-independent standard, with an honest 1/3-clean characterization of native-resolution single-band modeling on Euclid Q1 (P3, §8). The campaign follows the repository's retraction culture: goalposts (the verbatim thresholds of

[1] Source artifact: `research/notes/gigalens2-positioning.md`.

Table 10) were frozen before any science ran, and every number below is traced to a JSON artifact and a dated gate record in `CAMPAIGN.md`.

2 Data and Targets

2.1 The real system and its three drizzle products

We model DESI–165.4754−06.0423, observed with *HST*/WFC3-IR in F140W under program GO–15867 (public via MAST DOI 10.17909/hx0v-9260) and reduced by Huang et al. (2025) into three data products at different pixel scales, each with its own empirical PSF and heteroscedastic error map (Table 1). The scales differ by resampling: the native $0\overset{''}{.}128$ product (`v2d`, 80^2 px) is the pipeline drizzle at its true plate scale, the fine product (`v3`, $0\overset{''}{.}04$, 260^2 px) is a MAST HAP fine-skycell mosaic of the same exposures, and the binned product (`v3b`, $0\overset{''}{.}08$, 130^2 px) is a 2×2 rebuild of the fine mosaic. Because drizzle (Fruchter and Hook, 2002) distributes each input pixel's flux over an output footprint, the resampled products carry noise that is correlated on the drizzle-kernel scale (§3).[2] The model-subtracted residual autocorrelation confirms this directly: the along-axis lag-1 correlation is $\rho(1) = 0.815$ on the fine product, 0.615 binned, and 0.305 native.[3]

Table 1: The three drizzle products of DESI–165.4754−06.0423 and their measured noise correlation. Kept-pixel counts are after mask erosion by the whitening stencil (§3); $\rho(1)$ is the along-axis lag-1 correlation of the model-subtracted residual.

tag	scale ($''$)	grid	$\rho(1)$	kept px	role
`v3` (fine)	0.04	260^2	0.815	66,752	most correlated; artifact host
`v3b` (binned)	0.08	130^2	0.615	16,653	bimodal-γ host
`v2d` (native)	0.128	80^2	0.305	5,865	diagonal-trustworthy anchor

Note. Effective independent-pixel fraction $N_{\rm eff}/N \approx 0.017$; a 2×2 fine-pixel block is $\sim$78% internally correlated. The native product is where the diagonal likelihood is least wrong and anchors the cross-scale comparison (§6).

2.2 The drizzle-mock generator

Calibration with known truth requires mocks whose noise correlation is generated by the *same* drizzle operator that the likelihood models. Our generator (`05_gen_drizzle_mocks.py`) renders a noiseless lensed image on the fine grid, applies the analytic 3×3 separable tent convolution induced by the 3-frame drizzle, and adds correlated noise with the exact analytic covariance of each product. Machine-precision self-consistency is enforced: the noiseless render agrees with the drizzle operator to a worst deviation of $2.1 \times 10^{-12}\sigma$ over 8 mock trios, far inside the 0.05σ gate.[4]

One deviation is preserved deliberately as a real-data caveat.[5] A naïve 3-dither Latin stack $\{(0,0),(1,2),(2,1)\}$ is *provably shift-variant* — no single effective convolutional PSF exists, and the render check fails at 2.1–26.7σ. The mocks therefore use all nine dither phases, for which the effective PSF is exactly the separable 3×3 tent. The real `v3` skycell (`NDRIZIM`= 3) is likewise

[2] Source artifact: `data/noise_kernel_report.json`.

[3] These are the along-axis correlations of the detrended, model-subtracted residual measured for the kernel fit (`data/noise_kernel_report.json`); the $\rho(1) = 0.795/0.52$ fine/native figures quoted in the introduction and in Huang et al. (2025) are the isotropic sky-autocorrelation values of the raw product. Both express the same fact: the fine grid is heavily oversampled relative to its resolution element.

[4] Source artifact: `data/mocks_report.json`.

[5] Source artifact: `data/mocks_report_3frame.json`.

shift-variant: its "effective PSF" is an approximation, a genuine limitation of stationary-kernel modeling on real drizzle that we flag rather than hide.

2.3 The posterior zoo

For the sampler benchmark (§4) we freeze a four-tier "posterior zoo" of 19 targets that exercises distinct pathologies: analytic synthetic targets with known evidence and modes (T0), the twelve Gu et al. (2022) mock systems (T1), the 46-dimensional marginalized real posterior of DESI–165.4754−06.0423 at condition number $\sim 10^{14}$ (T2), and its 74-dimensional genuinely bimodal binned posterior (T3). The full specification is Appendix C. The Euclid Q1 targets (T4) (Euclid Collaboration et al., 2025) are the independent real-data test of the end-to-end phase (§8).

3 Methods I — The Correlated-Noise Likelihood

3.1 Covariance model and drizzle anchor

We factor the pixel covariance as

$$C \;=\; D^{1/2}\,K\,D^{1/2}, \tag{1}$$

where $D = \mathrm{diag}(\sigma_i^2)$ is the existing heteroscedastic error map (masks retained as $\sigma \to 10^{10}$) and K is a stationary correlation matrix, fixed per product, generated by a kernel $\rho(\Delta)$ fit on the *model-subtracted* residual autocorrelation (a guard-enforced requirement, so diffuse lens flux cannot bias the calibration). The kernel is anchored on the analytic drizzle correlation: for a square pixel of drizzle radius $r = 3.2075$ fine pixels the closed-form along-axis lag-1 correlation is $t(1) = 0.76805$, matched by the enumerated drizzle operator to 0.76799.[6]

Kernel family (deviation from the naive two-parameter plan). The pre-registered plan proposed a single-Gaussian family $\rho = (1-w)\delta + w\,[\rho_{\mathrm{drz}} \circledast G(\sigma_e)]$. This family cannot pass the $\max|\rho_{\mathrm{fit}} - \rho_{\mathrm{meas}}| \le 0.05$ gate on any product (best residuals 0.090/0.311/0.166 for `v2d`/`v3`/`v3b`); the residuals carry a medium-scale correlated pedestal, an anisotropic core, and, on `v2d`, column stripes. We adopt the minimal positive-semidefinite-by-construction extension — a two-component bivariate form,

$$\rho \;=\; (1 - w_d - w_b)\,\delta \;+\; w_d\,\big[\rho_{\mathrm{drz}} \circledast G_2(\sigma_{ey}, \sigma_{ex}, c_e)\big] \;+\; w_b\,G_2(\sigma_{by}, \sigma_{bx}, c_b), \tag{2}$$

whose PSD is a non-negative sum of a floor, the drizzle-anchored component, and a broad bivariate-Gaussian pedestal. It passes on all three products ($\max|\rho_{\mathrm{fit}} - \rho_{\mathrm{meas}}| = 0.0448/0.0270/0.0326$), with the single-family failure numbers preserved on record.[7] An independent cross-check validates the block-sum relation between scales: the binned kernel predicted from the fitted fine kernel agrees with the binned kernel measured on the block-summed fine residual to 0.031 (gate 0.05), confirming that our kernels commute correctly with 2×2 binning; the product-level disagreement (0.0595, informational) is the known non-commutation of the PHOTUTILS background detrend with binning.

3.2 Convolutional whitening

We whiten by a truncated convolution $h = \mathcal{F}^{-1}[S^{-1/2}]$, where $S(\omega)$ is the kernel PSD, regularized by a spectrum floor $s_{\mathrm{floor}} = 0.05$ (the critical conservative regularizer: the true fine-grid covariance

[6] Source artifact: `data/noise_kernel_report.json`.
[7] Source artifact: `data/noise_kernel_report.json`.

is near rank-deficient), truncated to a $(2M+1)^2$ stencil and Gauss–Newton refined. The operator is accepted only if its operator-norm whiteness gate

$$e_{\rm op} \;=\; \max_{\omega}\big|\, S(\omega)\,|\hat{h}_M(\omega)|^2 - 1\,\big| \;\le\; 0.02 \tag{3}$$

is met, *and* a Monte-Carlo whiteness check gives unit variance ($\mathrm{Var}(u) \in 1 \pm 0.02$) and lag-$\le$3 off-diagonal correlation below 0.01. Masks are handled by a *whiten-then-drop* rule: we keep only whitened pixels whose stencil avoids masks and the border (a `binary_erosion`), which is statistically exact. Table 2 summarizes the accepted whiteners. Two construction fixes were required in P1a: kernels are stored analytically out to half-width 64 to suppress truncation ringing, and the spectrum floor is *adaptive* (a hard 0.05 floor on an already-PSD kernel biases $\mathrm{Var}(u)$ to 0.981 and fails the gate).

Table 2: Accepted whiteners. $e_{\rm op}$ is the operator-norm whiteness residual (gate ≤ 0.02); MC $\mathrm{Var}(u)$ and worst lag-≤ 3 off-diagonal from 500–2000 draws (gates 1 ± 0.02, < 0.01). Pixel loss is mask/border erosion by the $(2M+1)^2$ stencil.

product	M	$e_{\rm op}$	$\mathrm{Var}(u)$	worst off-diag	kept / eroded
`v2d` (native, strict)	14	0.0177	1.004	0.0075	5,865 / 487
`v3` (fine)	20	0.0160	1.000	0.0003	66,752 / 37,519
`v3b` (binned)	10	0.0124	0.999	0.0012	16,653 / 9,273
`v2d_relaxed` (native)	10	0.0312	1.001	0.0029	5,865 / 1,466

Note. The strict `v2d` whitener erodes to only 487 kept pixels (91.7% loss from the 29^2 stencil at native scale). A relaxed whitener ($e_{\rm op} \le 0.05$, $M{=}10$, 1466 kept = 3.0× strict) was built as a pre-registered evidence-driven exception and adopted after mock calibration held (§5). Source: `data/whitener_report.json`.

3.3 Generalized ridge marginalization

The 28 shapelet source amplitudes are linear parameters; we marginalize them analytically using their Gaussian priors as a Tikhonov ridge, generalizing the diagonal-likelihood marginalization of Huang et al. (2025) to the whitened residual. Writing $\tilde{X} = W_e[\,D^{-1/2} \circledast h\,]X$ for the whitened, PSF-convolved unit-amplitude design matrix (W_e the keep-mask, $\circledast h$ the whitening convolution) and $\tilde{R}$ for the whitened residual after the deterministic components,

$$A = \tilde{X}^\top \tilde{X} + \Lambda, \qquad b = \tilde{X}^\top \tilde{R}, \qquad a^\star = A^{-1} b, \tag{4}$$

$$\log L = -\tfrac{1}{2}\|\tilde{R}\|^2 + \tfrac{1}{2}\, b^\top a^\star - \tfrac{1}{2}\log\det A + \mathrm{const}, \tag{5}$$

with $\Lambda_{ii} = (i+1)/25$ the prior precision (so $A \succ 0$ and the solve is Cholesky, never a pseudo-inverse), and $-\frac{1}{2}\log\det A$ the Gaussian-evidence (Occam) term. The additive constant absorbs $-\frac{1}{2}\log\det C$, which is θ-independent (the kernels are fixed) and therefore dropped for sampling but *stored*: cross-whitener evidence comparisons must add it back, and the exact-versus-Szegő constant gap is +27.30 (`v2d`) and +179.21 (`v3b`) nat.[8]

3.4 Exact-reference validation and the grouped-convolution fix

Two independent references certify the implementation. First, the *diagonal limit*: with $C = D$ the whitened functional (4)–(5) must reduce *exactly* to the validated diagonal stack. It does, to

[8] Source artifact: `data/exact_ref_report.json`.

$|\Delta \log L| = 0$ at machine precision across four test points, and the whole marginalized log-posterior and its gradient reproduce the Foundry-I `_hmc_lib_marg` core to $|\Delta \log p| = 0$ and relative gradient $L_2 = 7 \times 10^{-17}$ (Appendix A).[9] Second, the *dense-covariance* reference: the convolution-whitened $\log L$ is compared against a CPU float-64 dense-C Cholesky on the kept pixels at 20 prior draws. The worst $|\Delta \log L|$ is 2.79×10^{-9} nat (`v2d`) and 6.26×10^{-7} nat (`v3b`), both far inside the 0.1-nat gate; on the 260^2 fine grid an FFT-preconditioned conjugate-gradient solve (relative residual 10^{-8}, 131 iterations) reproduces the exact GLS quadratic form.[10] We stress the gate's semantics: it certifies *implementation-exactness* of the whitened functional $C^{-1} := G_e^\top G_e$; physical-C misspecification is bounded separately by e_{op} (Eq. 3), since exact-GLS equivalence at individual prior draws is mathematically unattainable under any finite floor.

The batched gradient exposed a genuine stack defect (§10, #7): a `vmap` over chains of 28 per-column convolutional whitening operations *livelocks* the XLA compiler (100% CPU, GPU idle, >20 min), which disabling priority-fusion alone does not cure. Collapsing the 28 per-column convolutions into a single depthwise / grouped convolution makes the batched gradient compile in 13.8 s and leaves the log-posterior bit-identical. The whitened correlated gradient then costs $\leq 1.6\times$ the diagonal one (40/85 ms versus 25/55 ms on an L4), so the campaign runs at diagonal-comparable throughput.

4 Methods II — The Posterior Zoo and Benchmark Protocol

4.1 The `LensPosterior` interface

Every zoo target exposes a single frozen, jitted target through a `LensPosterior` dataclass. Its defining feature is a *`log_prior` / `log_like` split in z-space*, which enables likelihood-only tempering (for SMC, parallel tempering, and thermodynamic integration) and evidence estimation, alongside the bijector with explicit leaf-order labels (solving the Gu et al. (2022) parameter-ordering trap once), a unit-cube face for NAUTILUS, and an `InitBundle` of MAP/floored-SVI/prior starts. Adapter fidelity is enforced: each sampler adapter asserts `log_prob` equality against the frozen target at three reference points before any benchmark run, so no two contenders can silently sample different posteriors. The freeze is content-addressed (`sha256` per target) and re-checked bit-identically across processes and architectures (A16↔L4, float-64).[11]

4.2 Targets and validation gates

The 19 targets and their five acceptance gates (T1 bit-identity, T2 stored- logp reproduction, T3 basin structure, T0 analytics, and prior+like=prob consistency) are specified in Appendix C; all five passed. Two values anchor the science: the T2 target reproduces the stored marginalized log-posterior to $\log p = -45840.984005998456$ (tolerance 10^{-8}), and the T3 target reproduces the disconnected binned basins exactly — 45 low-slope chains ($\bar{\gamma} = 1.294$) and 3 steep chains, occupancy 0.9375/0.0625, with zero migrations.

4.3 Contenders

Nine contenders are benchmarked (Appendix D): the GIGA-LENS baseline (S0: multi-start MAP → SVI → ChEES-HMC); window-adapted NUTS and unadjusted MCLMC (BLACKJAX); FLOWMC

[9] Source artifact: `data/parity_report.json`.
[10] Source artifact: `data/exact_ref_report.json`.
[11] Source artifact: `data/zoo_validation.json`.

`RQSpline_MALA`; adaptive tempered SMC (BlackJAX); nautilus (Lange, 2023); parallel- tempered HMC (TFP replica-exchange over our batched preconditioned HMC, which also generates the T3 reference); NeuTra-HMC via a normalizing-flow pullback (Hoffman et al., 2019); and the ranked recipe bet "GL-NT" — MAP → SVI → short tempered-SMC anneal → neural-spline-flow fit → ChEES-HMC in flow space. pocoMC (Karamanis et al., 2022) was a pre-approved drop (adaptive tempered SMC covers the preconditioned-SMC science).

4.4 Protocol: budget-matched, until-converged, pre-registered

Two tracks quantify complementary questions. **Track A** is *budget-matched*: every contender receives an identical gradient-evaluation budget (2×10^5 for T0, 1.5×10^6 for T1, plus a 692k init-cache billed identically to all consumers; gradient-free samplers get a 2×-likelihood budget), and we compare median-over-three-seeds ESS per gradient. **Track B** runs *until converged* ($\hat{R} \leq 1.01 \wedge \mathrm{ESS}_{\min} \geq 1000$, capped at 4× the Track-A budget) and reports the cost to reach convergence. The win criterion, frozen in advance, is: median ESS/gradient $\geq 2\times$ baseline *and* rank-normalized $\hat{R} < 1.01$ *and* (multimodal targets) every reference mode recovered with occupancy error ≤ 0.10.

Decisively, all hyperparameter policies were *frozen on a development split* (all T0 plus Gu et al. (2022) systems 0–5, seed 0; ≤ 8 configs per method, 33^+ logged trials) and git-timestamped *before any evaluation cell was read* (`data/policies_frozen.json`, `2026-07-06T22:08Z`).[12] No policy moved for the evaluation runs. This policy-freeze-before-eval discipline is what lets §7 report a benchmark rather than a tuning exercise, and its generalization is measured directly (Table 6).

5 Mock Validation Results

We validate the correlated likelihood on drizzle mocks with known truth in four experiments (E1a–E1d); the outcomes are Table 3, and the pooled diagnostics are Figures 1–3. The headline of this section is not only that the correlated likelihood calibrates, but that reaching that verdict required — and produced — a sampler-recipe finding of independent value.

5.1 E1a: the artifact, reproduced with known truth

Applying the *diagonal* likelihood to fine-product mocks whose noise is genuinely drizzle-correlated reproduces the real-data pathology in miniature: the recovered slope is biased at median $|z(\gamma)| = 5.84$ with 68% coverage of exactly 0 over eight mocks — the motivating effect, demonstrated against known truth.[13] This is the mock-space analogue of the fine-product $\gamma = 2.585$ artifact on the real system.

5.2 The two-stage PHMC finding (sampler root cause)

The first pass of the recovery experiment (E1b) *appeared* to fail: under the correlated likelihood the pooled bias was $\bar{z}(\gamma) = -0.65$ (fine) and $+1.22$ (binned), outside the $|\bar{z}| < 0.5$ gate. Rather than move the goalpost, a pre-registered diagnosis pass established that the outlier z values coincided exactly with *unhealthy fits* ($\hat{R}$ up to 2.1, minimum ESS 30–240 at reduced budget): the gate was confounded by sampler depth, not by the likelihood. The root cause is that a floored-SVI covariance is too poor a momentum metric for these near-degenerate 22-dimensional posteriors (tripling depth alone still gave $\hat{R} = 3.11$ on a canary mock). The fix, now the production recipe, is a *two-stage*

[12] Source artifact: `data/policy_tuning_log.json`.

[13] Source artifact: `data/e1_report.json`.

preconditioned HMC: a first stage of 24 chains draws from the floored-SVI metric, the momentum metric is then re-estimated from the *pooled cross-chain* stage-1 draws, and a second stage warm-starts from the stage-1 chain ends. On the canary this lowers $\hat{R}$ from 3.11 (depth-only) to 1.003 with ESS 13k–22k, at $\sim$1.9× cost; 63/63 deep re-runs are clean.[14] That a modest change to metric estimation — not to the model — turns an apparent calibration failure into a pass is itself a contribution, and it is the recipe carried into the real-data phase (§6).

5.3 E1b/E1c/E1d under the production recipe

E1b (depth-controlled) passes at every scale. The pooled slope bias is $\bar{z}(\gamma) = -0.36$ (fine), -0.33 (binned), -0.05 (native), all inside $|\bar{z}| < 0.5$, and the cross-scale agreement gate passes 7/8 (each fine posterior agreeing with its native counterpart within $1\sigma_{\mathrm{comb}}$).

The width ratio is a characterized cost, not a bias. The one gate E1b does not meet is the posterior-width ratio: median $\sigma_{\mathrm{fine}}/\sigma_{\mathrm{native}} = 2.45$, outside $[0.7, 1.5]$. Because the *bias* and *cross-scale* gates pass, this is diagnosed as *conservative, not biased*: the δ-regularized fine whitener discards information at the tent kernel's spectral zeros, so the fine posterior is wider than the same-photon native posterior. This is a genuine, honestly reported cost of near-singular drizzle covariances; a λ-sensitivity arm was added to the robustness phase (pre-registered before the real-data runs), and the exact-GLS information loss is quantified in §3.4.

E1c (SBC-lite) passes on healthy fits. Over all 64 mocks the pooled rank test flags γ ($p = 6.5 \times 10^{-5}$; U-shaped ranks) — the signature of stuck chains, not miscalibration. Restricted to the 13 healthy fits ($\hat{R} < 1.05$, ESS> 100), all six mass parameters pass rank uniformity (γ: $p = 0.53$) and the pooled 68% coverage is $0.615 \in [0.55, 0.80]$ (low-n caveat noted; a definitive full-64 staged re-run is queued, non-gating).

E1d: kernels exonerated, relaxed whitener adopted. A pre-registered kernel-attribution arm (D2) compares the fitted kernel against an analytic drizzle kernel: both calibrate at depth (healthy $\bar{z}(\gamma) = -0.14$ and -0.47), so the original failures were sampler depth, *not* kernel fitting or δ-regularization. Arbitrating strict versus relaxed native whiteners (D3), the relaxed whitener is adopted — $\max|\bar{z}| = 0.492$, pooled coverage 0.594, keeping 4.9× as many pixels as the strict one, with the strict whitener also calibrating ($\max|\bar{z}| = 0.469$). Critically, the diagonal likelihood *fails* even at native scale under the real `v2d` correlation kernel ($\bar{z}(\gamma) = -6.1$): the artifact is strong even where the diagonal likelihood is least wrong.

Table 3: Mock-validation gate outcomes (E1) under the two-stage PHMC production recipe. All numbers: `data/e1_report.json`.

gate	pre-registered threshold	measured	verdict
E1a artifact	median $\lvert z(\gamma)\rvert > 2$ OR cov$< 40\%$	5.84; cov 0%	**PASS** (artifact repr
E1b bias	$\lvert\bar{z}(\gamma)\rvert < 0.5$ / scale	$-0.36/-0.33/-0.05$	**PASS** (all scales)
E1b cross-scale	$\geq 6/8$ within $1\sigma_{\mathrm{comb}}$	7/8	**PASS**
E1b width ratio	$\sigma_{\mathrm{fine}}/\sigma_{\mathrm{nat}} \in [0.7, 1.5]$	2.45	FAIL (conservative,
E1c rank (6 params, healthy)	$p > 0.01$ each	γ $p = 0.53$	**PASS**
E1c coverage (healthy)	pooled 68% $\in [55, 80]\%$	61.5%	**PASS**
E1d D2 kernels	both arms calibrate	$\bar{z} = -0.14, -0.47$	kernels exonerated
E1d D3 whitener	relaxed if calibrated & $\geq 3\times$ px	relaxed adopted	**PASS**

[14] Source artifact: `data/e1_report.json`.

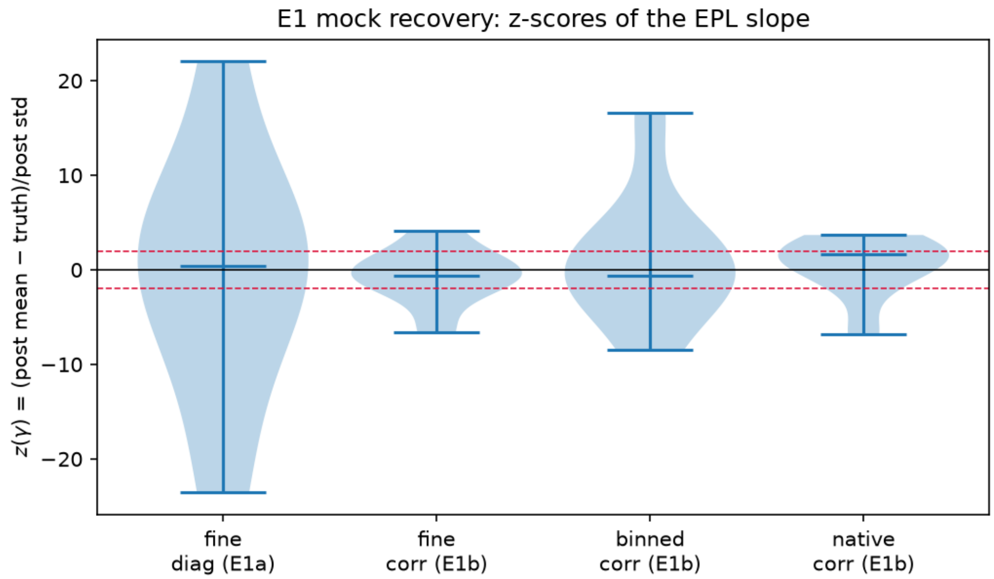

Figure 1: Recovered-slope z-score distributions across mocks, by product scale and likelihood. The diagonal likelihood on drizzle-correlated fine mocks (the E1a artifact) produces the broad, biased violin; the correlated likelihood under the two-stage recipe centers each scale within the $|\bar{z}| < 0.5$ band. Source: `08_pool_e1.py`.

6 Real-Data Cross-Scale Unification and the Bimodality Verdict

This is the campaign's central real-data result. The headline is one sentence: the correlated-noise likelihood is **necessary but not sufficient**. *It removes the diagonal likelihood's upsampling artifact — and, decisively, flips the binned bimodality verdict — but it over-corrects the slope value, so the scales bracket the anchor rather than unifying onto it.*

The correlated likelihood is applied to all three real products of DESI–165.4754−06.0423 with multi-basin starts, testing three pre-registered hypotheses (both branches of each written down in advance; no goalpost moves). The native fit uses the D3-adopted relaxed whitener; a pre-registered amendment (recorded before any converged run) anchors H2/H3 on the *diagonal*-native fit $\gamma = 1.433\,[1.400, 1.469]$, because the relaxed-whitener native correlated fit is likelihood-weak (information-limited by pixel loss, the real-data analogue of the E1b width finding) and is prior-pulled toward $\gamma \approx 2.0$; its width is reported separately as a pixel-loss information-cost diagnostic. The money comparison is therefore whether the correlated likelihood pulls the scale-dependent slopes back onto the trustworthy native anchor. Table 4 and Figure 4 collect the outcome.

6.1 The cross-scale slope and the money number

The correlated likelihood does move each product's slope substantially and in the pillar's predicted direction, but not onto a single value. On the fine (`v3`, 37,519 px) product the steep basin deflates from the diagonal artifact $\gamma = 2.585$ to $\gamma = 1.816\,[1.703, 1.930]$ ($\hat{R} = 1.03$, $\mathrm{ESS}_\gamma = 2047$) — a $\Delta = 0.77$ correction, and, critically, *below* the γ prior mean 2.0, so the information-rich fine likelihood pulls the slope down data-driven rather than by prior. On the binned (`v3b`, 9,273 px) product, the low basin — the **money number** — converges to

$$\gamma^{\mathrm{binned}}(\mathrm{corr,\ low}) \;=\; 1.103 \pm 0.008 \quad (q_{16/84} = 1.096/1.112,\ \theta_{\mathrm{E}} = 2.624 \pm 0.005),$$

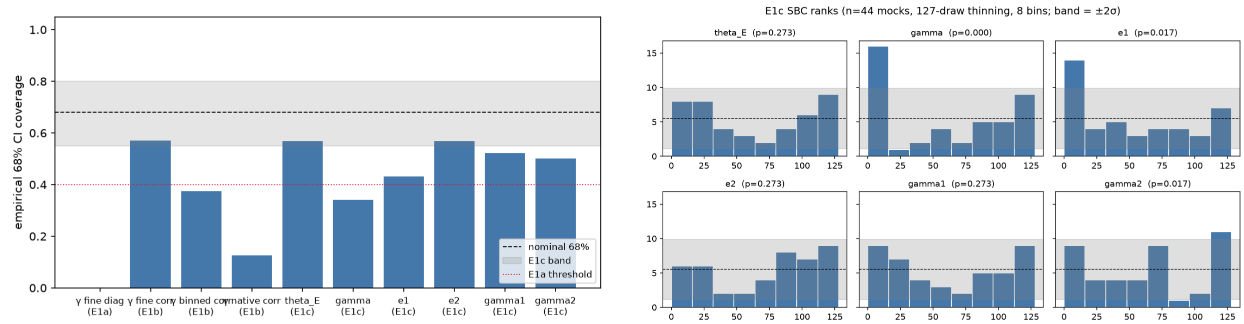

Figure 2: Left: empirical 68% coverage per parameter (healthy fits), with the $[55, 80]\%$ acceptance band. Right: SBC rank histograms; γ is uniform on healthy fits, and the U-shape that flagged it over the full set is the signature of under-mixed chains, not miscalibration (§5.2).

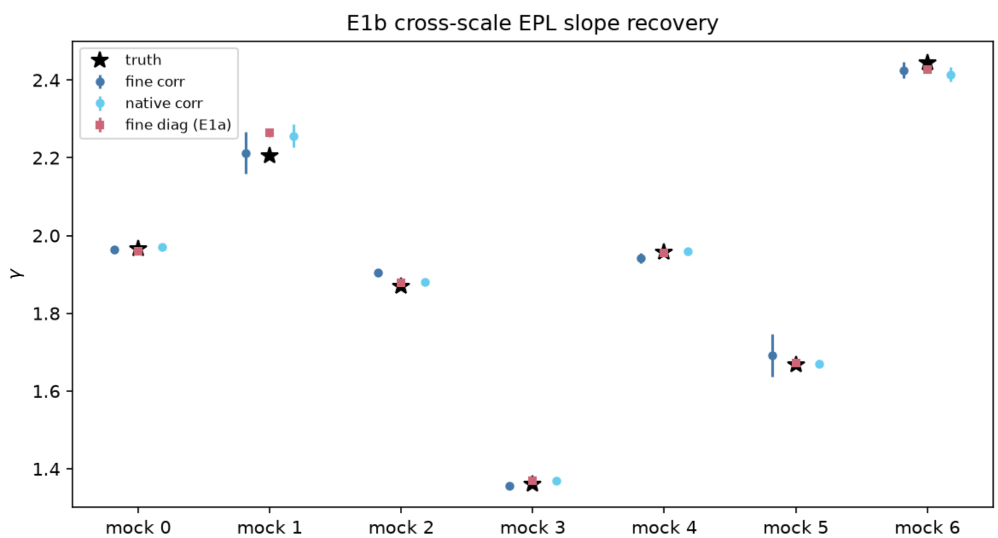

Figure 3: Cross-scale slope agreement on mocks: fine versus native γ under the correlated likelihood, with combined 1σ error bars. Seven of eight pairs agree within $1\sigma_{\text{comb}}$; the fine posteriors are systematically wider (the characterized $\sigma_{\text{fine}}/\sigma_{\text{native}} = 2.45$ information cost, not a bias).

sampled by a 128-particle tempered SMC (Sec. 6.3). This value is $\sim 17\sigma$ *below* the diagonal-native anchor 1.433.[15] The correlated-native fit itself provides no help: it drifts to $\gamma \approx 2.35$ (relaxed whitener, both basins), a prior-pulled over-whitening artifact of the information-poor native product, which is exactly why the pre-registered amendment anchors on the diagonal-native value. Together the correlated slopes *bracket* the anchor — fine-steep 1.816 above (3.1σ high), binned-low 1.103 below, native prior-pulled — but do not converge to it.

The pre-registered hypotheses, resolved.

H1 — bimodality: *CONFIRMED (artifact branch).* Under the correlated likelihood the binned steep basin carries evidence weight $w_{\text{steep}} \approx 0$: per-basin SMC gives $\Delta \log Z(\text{steep} - \text{low}) = -28.9$ nats, favouring low. This flips the diagonal likelihood's +162-nat steep preference (Sec. 6.2). The `v3b` bimodality is a noise-covariance / upsampling artifact, not a genuine

[15]The $\sim 17\sigma$ figure is the ledger value (`CAMPAIGN.md`, P1c MONEY 2026-07-14), combining the SMC width 0.008 with the anchor uncertainty; even against the anchor's own $\sigma_\gamma \approx 0.034$ the discrepancy is $\sim 9.5\sigma$.

Table 4: Cross-scale power-law slope γ of DESI–165.4754−06.0423 under the diagonal and correlated likelihoods, by product and basin. The diagonal-native fit is the pre-registered anchor. The money number (bold) is the correlated binned-low slope from 128-particle tempered SMC. Basin weights w are per-basin SMC evidence (Table 5). Diagonal references, fine-steep, and native-relaxed: `data/e2_report_interim_unconverged.json`; converged binned-low/steep SMC: `CAMPAIGN.md` (P1c MONEY, 2026-07-14).

product (scale)	likelihood	basin	γ (68% CI)	note
`v2d` 0.128″	diagonal	–	$1.433\,[1.400, 1.469]$	**anchor** (σ_γ=0.034)
`v2d` (relaxed)	correlated	low	$2.353\,[2.258, 2.450]$	prior-pulled, info-poor
`v3b` 0.08″	diagonal	low	1.293 ± 0.012	$w_{\rm low} \approx 0$ (evidence)
`v3b`	diagonal	steep	2.423 ± 0.027	$w_{\rm steep} \approx 1$ ($\Delta \log Z$=+162)
`v3b`	correlated	low	$\mathbf{1.103}\,[\mathbf{1.096}, \mathbf{1.112}]$	**money**; $w_{\rm low} \approx 1$ ($\Delta \log Z$=−29)
`v3b`	correlated	steep	2.64	$w_{\rm steep} \approx 0$
`v3` 0.04″	diagonal	–	2.585 (MAP)	upsampling artifact
`v3`	correlated	steep	$1.816\,[1.703, 1.930]$	deflated; $\hat{R}$=1.03

multimodality — the pre-registered artifact branch, established rigorously by evidence rather than by chain occupancy.

H2 — cross-scale unification: *REJECTED (over-correction).* The binned-low slope 1.103 ± 0.008 is $\sim 17\sigma$ below the anchor and the fine-steep slope 1.816 is 3.1σ above it; $|\Delta\gamma| \gg 2\sigma_{\rm comb}$ at both scales, and the 68% intervals do not overlap. The correlated slopes bracket the anchor rather than unifying onto it: the noise model is a large, real correction but leaves a residual scale dependence (Sec. 6.5).

H3 — honesty: *PASS.* The fine correlated posterior is appropriately wide: $\sigma_\gamma(\text{fine, corr}) = 0.117 \geq \sigma_\gamma(\text{native, diag})/1.5 = 0.034/1.5 = 0.023$. The same photons are not made to manufacture information — the information-cost characterized on mocks (§5) carries over honestly to the real data.

6.2 The bimodality verdict: a 191-nat evidence flip

The sharpest single result of Pillar 1 is that the correlated likelihood *flips the sign* of the binned basin evidence (Fig. 5, Table 5). Under the *diagonal* likelihood, per-basin tempered-SMC evidence on the 74-dimensional binned posterior favours the *steep* basin decisively — $\log Z_{\rm steep} - \log Z_{\rm low} = +162.2 \pm 1.8$ nats ($w_{\rm steep} \approx 1$) — exposing the stored-chain 45-low/3-steep occupancy split (naive $w_{\rm low} = 0.94$) as a start artifact with zero inter-basin migrations. Under the *correlated* likelihood the same comparison gives $\log Z_{\rm steep} - \log Z_{\rm low} = -28.9$ nats ($w_{\rm low} \approx 1$): the evidence swings by $\sim$191 nats from steep to low. Because the two likelihoods carry different (fixed) whitening log-determinant constants, only the *within*-likelihood $\Delta \log Z$ is comparable across the flip; that difference is what matters, and it is unambiguous. The steep basin's diagonal dominance on the upsampled product was a noise-covariance artifact, and the correlated likelihood removes it — the cleanest confirmation that the drizzle correlation, not the physics, produced the diagonal likelihood's apparent multimodality.

6.3 Why sampling this posterior required tempered SMC

Extracting the money number was not routine, and the difficulty is itself a finding that ties Pillar 1 to the sampler benchmark (§7). The two-stage PHMC recipe that calibrates the mocks (§5.2) *fails*

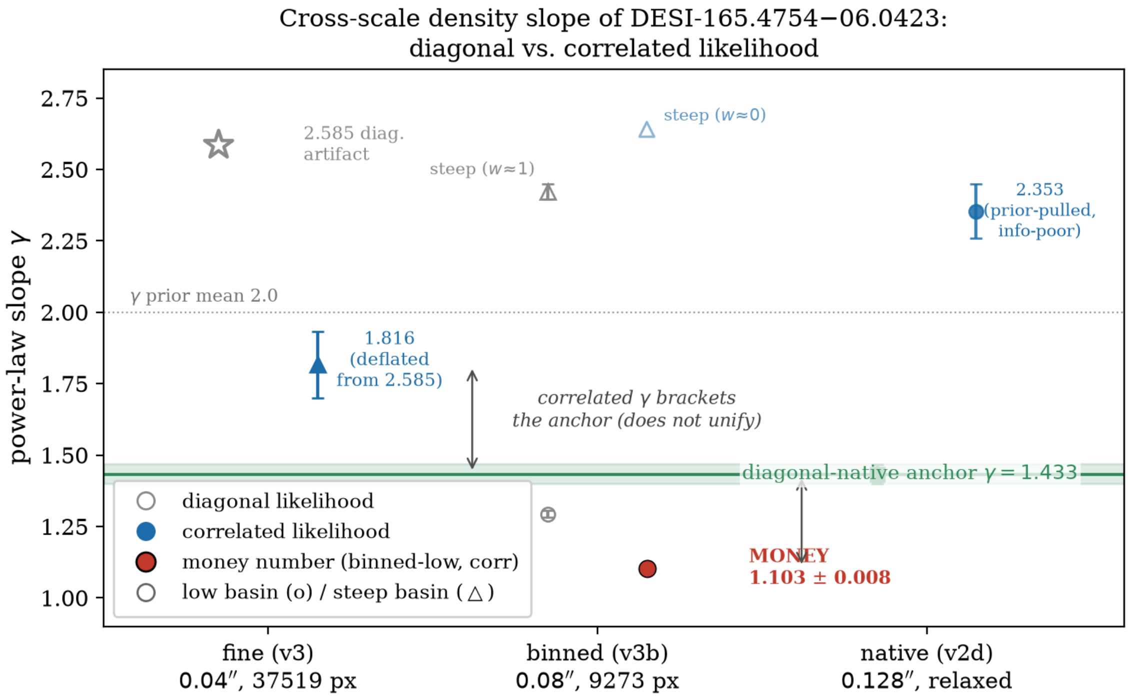

Figure 4: (Headline figure.) Cross-scale slope γ of DESI–165.4754−06.0423: diagonal (grey, open) versus correlated (blue/red, filled) likelihood at the fine, binned, and native scales, with 68% intervals and the diagonal-native anchor band 1.433 [1.400, 1.469] (green). Circles are the low basin, triangles the steep basin; the red point is the money number, the correlated binned-low $\gamma = 1.103 \pm 0.008$. The correlated likelihood deflates every diagonal slope toward the anchor (fine $2.585 \to 1.816$; binned steep-dominated $\to$ low 1.103) but *brackets* rather than unifies onto it. Source: committed P1c results, plotted by `make_p1c_figs.py`.

on the real binned-low correlated posterior. The root cause is that the correlated-likelihood MAP is a *saddle, not a mode*: its Laplace Hessian has minimum eigenvalue −14.85 with five negative directions, and the highest- density point ($\gamma_{\text{best}} \approx 1.10$, $\log p = -4683$) lies away from the polished MAP ($\gamma_{\text{map}} = 1.27$, $\log p = -4757$). A momentum metric built from a saddle Hessian is ill-posed; both principled positive-definite metric repairs we tried (a diagonal $|H_{ii}|$ metric and a full-rank SVI covariance) leave $\hat{R}$ at 21–22, and the two-stage re-preconditioning makes it *worse* by pooling unmixed stage-1 draws. Crucially, the $\hat{R} \sim 22$ is a nuisance, not the slope: it is carried by a Sérsic-versus-shapelet source-*centre* degeneracy (two clusters at −0.15 and +0.09, source-centre $\hat{R} = 22.3/15.1$) that is decoupled from γ. A 128-particle tempered SMC (annealing $\lambda \to 1$ in 28 steps, ESS = 77–118) crosses those source-centre sub-modes where a single HMC mass matrix cannot, and returns both the γ marginal with a proper σ and the basin evidence of Sec. 6.2 in one tool. Convergence tracks Laplace-Hessian definiteness across all products: the fine steep basin (min eigenvalue +0.11, positive-definite) mixes cleanly under PHMC ($\hat{R} = 1.03$), whereas the indefinite binned and native targets freeze. The two-stage recipe was validated on positive-definite mock Hessians and does *not* transfer to these real-data saddle targets — the same lesson the T2/T3 rows of the benchmark reach from the sampler side.

Table 5: Per-basin SMC evidence on the binned `v3b` posterior under the two likelihoods. Only the *within*-row $\Delta \log Z$ is comparable (the absolute $\log Z$ carry different whitening constants). Diagonal: 300-particle SMC (`CAMPAIGN.md`, P2c partial #2a; `data/results/basin_evidence/`). Correlated: 128-particle SMC (P1c MONEY, 2026-07-14).

likelihood	$\log Z_{\rm low}$	$\log Z_{\rm steep}$	$\Delta \log Z_{\rm steep-low}$	verdict
diagonal	38351.17 ± 0.66	38513.37 ± 1.71	$+162.2 \pm 1.8$	steep ($w_{\rm steep} \approx 1$)
correlated	-4771.08	-4799.96	-28.9	low ($w_{\rm low} \approx 1$)

Swing: $+162 \to -29 =$ **191** nats, steep $\to$ low.

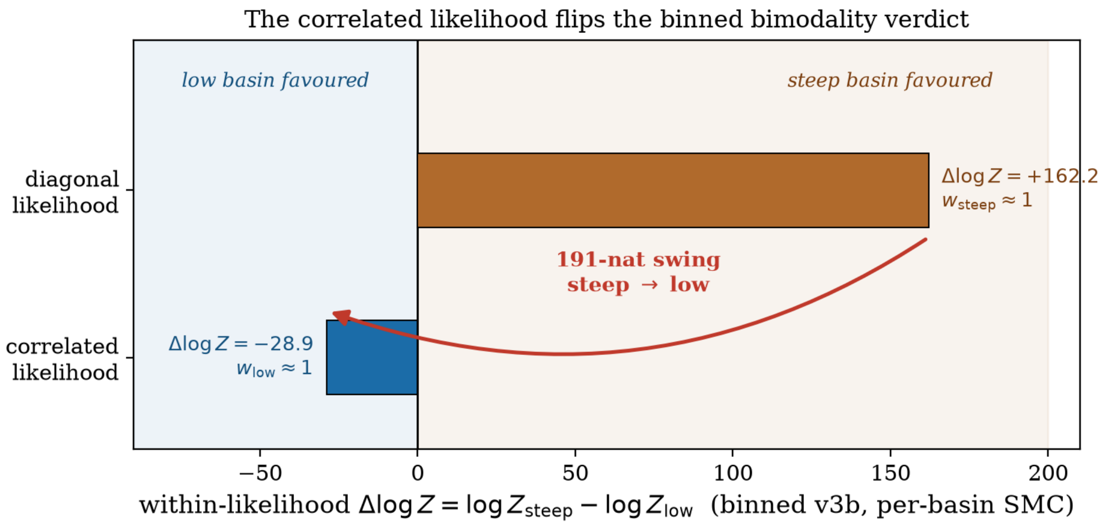

Figure 5: The correlated likelihood flips the binned bimodality verdict. The within-likelihood evidence difference $\Delta \log Z = \log Z_{\rm steep} - \log Z_{\rm low}$ (per-basin SMC on `v3b`) moves from $+162.2$ (diagonal, steep-favoured) to -28.9 (correlated, low-favoured) — a 191-nat swing. The diagonal likelihood's steep-basin dominance on the 2×-upsampled product is a noise-covariance artifact the correlated likelihood removes (H1 confirmed). The stored-chain 45/3 occupancy is a start artifact (0 migrations); evidence, not occupancy, is what weights the modes. Source: `make_p1c_figs.py`.

6.4 Whitener information cost and the excluded fine-low basin

Two honest limitations bound the result. First, the fine *low* basin is **excluded** as a characterized whitener pathology, not sampled. The near-singular fine (3.2×-upsampled) δ-regularized whitener can be *gamed*: two independent, physically sane starts (from $\gamma = 1.02$ and from the binned-consistent 1.28) both rail toward $\gamma \approx 1.0$, where the real-space $\chi^2_{\rm pp}$ climbs to 72–80 (a garbage image) even as the whitened log-likelihood *improves* — the fine low basin is unconstrained. This is the E1b spectral-zeros information loss ($\sigma_{\rm fine}/\sigma_{\rm native} = 2.45$, §5) surfacing as an outright MAP pathology; we therefore report the fine product steep-only and make the binned product the defensible cross-scale headline. Second, the native correlated fit is information-poor: the strict 487-pixel whitener is *unconvergeable* (Laplace minimum eigenvalue -1.3×10^{23}, 18/46 eigenvalues floored — below $\sim$500 kept pixels the whitened likelihood is near-flat), and even the relaxed 1466-pixel whitener is prior-pulled to $\gamma \approx 2.35$ under a TruncatedNormal$(2.0, 0.25)$ slope prior. Native drizzle is already only weakly correlated ($\rho(1) \approx 0.3$–0.5), so whitening it discards information for little benefit — confirming the pre-registered decision to anchor on the diagonal-native fit.

6.5 Necessary but not sufficient

The complete, honest verdict is nuanced and publishable in both its parts. On the *selection* of basins (H1), the correlated likelihood **wins**: it converts the diagonal likelihood's spurious +162-nat steep dominance to a −29-nat low preference, demonstrating that the binned bimodality is a noise-covariance artifact the correct likelihood removes. On the *value* of the slope (H2), it **fails by over-correction**: the restored low-basin $\gamma = 1.103 \pm 0.008$ lands $\sim 17\sigma$ below the native anchor, and the correlated slopes across scales bracket the anchor (1.816 fine-steep above, 1.103 binned-low below, $\sim$2.35 native prior-pulled) instead of collapsing onto it. The correlated noise model is therefore a major, real correction — it removes the diagonal likelihood's upsampling steep artifact (binned $\sim$100% steep $\gamma \approx 2.4 \to 0\%$ steep, γ deflated to 1.10; fine $2.585 \to 1.816$) — but it is *not sufficient* for clean cross-scale unification. The residual scale dependence points to additional systematics, most plausibly the source model and PSF (the very ingredients that the code-comparison literature (Galan et al., 2024) and the Euclid analysis of §8 independently implicate), which the shapelet source and stationary-kernel PSF of this campaign do not yet address.

7 Sampler Benchmark Results

The benchmark comprises 370 evaluation cells (Track A 345/351, the six omissions being pre-registered structured failures, plus 24 Track B cells), consuming 202.4 phoenix GPU-h and zero Perlmutter budget.[16] The T0/T1 story is complete and settled below; the until-converged story on the hardest real targets (T2 cond-10^{14}, T3 bimodal) is the A100 phase P2c, reported in §7.7. Figure 6 is the full efficiency matrix.

7.1 nautilus dominates efficiency — but is disqualified where it matters most

NAUTILUS (Lange, 2023) dominates sampling efficiency wherever it applies: its median ESS/gradient beats the GIGA-LENS baseline by 2.6–307× across the zoo (e.g. 180.7× on the hard Gu et al. (2022) system 3, 307× on system 11, 94× on the funnel, 114× on the well-separated mixture), it returns the best evidence estimates, and it recovers mixture modes (Table 7). *But* it is *disqualified* on the condition-number-10^{14} regime that the real marginalized lens posterior actually occupies: on `t0_illcond46` its live-point shells collapse (fewer than eight equal-weight points in 74 minutes) and its evidence estimate errs by $\sim 10^7$ nat. Its reported ESS is moreover importance-weight ESS with a weak pseudo-$\hat{R}$ on exchangeable draws — a semantics caveat we record rather than paper over: we do not fail NAUTILUS on the $\hat{R}$ criterion, but its mode-occupancy and evidence carry its weight, and neither survives cond-10^{14}.

7.2 At budget parity, no gradient method beats the baseline on T1

The sharpest budget-matched result is negative and important: *under Track-A budget parity, no gradient-based contender beats the GIGA-LENS baseline (S0) on the evaluation T1 systems.* The best gradient challenger, unadjusted MCLMC with an SVI-diagonal metric, reaches at most 1.3× the baseline ESS/gradient (system 1) and a median below 1×; NUTS, tempered SMC, NeuTra, and the GL-NT recipe are all well below parity on T1. Efficiency dominance on T1 belongs to NAUTILUS alone — and NAUTILUS is the sampler that fails the real marginalization regime. The reliable-but-not-faster methods (S0, PT-HMC) and the fast-but-fragile method (NAUTILUS) thus map onto different columns of the same matrix, which is the benchmark's central lesson.

[16] Source artifact: `data/bench_report.json`.

7.3 Until converged, reliability inverts toward PT-HMC

The Track-B (until-converged) ranking inverts the efficiency ranking. Parallel-tempered HMC (`remc_pt`) converges 5 of 6 hard T1 systems, *including* system 6, where the baseline does not converge; it fails only system 11 (where the baseline also fails). By contrast the baseline uniquely *owns* the cond-10^{14} regime: on `t0_illcond46` S0 is the only healthy gradient method ($\hat{R} = 1.00$) besides MCLMC ($\hat{R} = 1.04$), while PT-HMC blows up ($\hat{R} \sim 10^{16}$ with its Track-A β-ladder), NUTS reaches $\hat{R} = 3.76$, and NeuTra $\hat{R} = 5.72$.

7.4 flowMC fails structurally; GL-NT fails its own bars at Track-A budget

FLOWMC (Wong et al., 2023) (version 0.4.5) is a *structural* failure on lens posteriors, not a tuning miss: its scalar-MALA local step has no per-dim preconditioning, so on the ill-conditioned z-space it holds $\hat{R} = 1.9$–2.1 across T1, converges 0/6 Track-B systems even at 4× budget, and *inverts* the mixture on `t0_mix22` (occupancy 0.13/0.87 versus the true 0.8/0.2). This is a benchmark datum on the limits of flow-global-move sampling without local preconditioning.

The ranked recipe bet, GL-NT, *fails its own recipe bars at Track-A budget*: its T1-hard ESS/gradient is 0.03–0.05× the baseline (target $\geq 3\times$), it regresses on easy targets, and its rank-normalized $\hat{R} = 1.4$–2.0 — flow-space ChEES is the weak stage. Its mode recovery is real but partial (it clears the 0.05 mode-weight bar only in float-64 on the separated mixture) and never unique. GL-NT is therefore the lowest-priority P2c item: flow-space HMC needs re-tuning or an A100 budget before any "recipe" claim is admissible; per the pre-registered criteria, the current evidence supports a *benchmark* paper, not a *recipe* paper.

7.5 Multimodality: who recovers the modes

On the two-component mixtures the baseline, NUTS, and NeuTra collapse to a single mode (1.000/0.000 on `t0_mix2`); NAUTILUS, tempered SMC, FLOWMC, and PT-HMC recover both. On the *harder* well-separated mixture `t0_mix22`, only PT-HMC clears the < 0.05 mode-weight bar (median error 0.031), while FLOWMC inverts the weights. Mode-recovery reliability and sampling efficiency are, again, different axes: the efficient methods are not the reliable mode-finders and vice versa.

7.6 The freeze generalizes (no dev→eval overfit)

Table 6 closes the protocol loop. The evaluation-to-development ESS/gradient ratios are ∼1.0 for every contender (NAUTILUS 1.10, MCLMC 1.04, FLOWMC 1.03, GL-NT 1.03), i.e. frozen policies do *not* overfit the development split; the persistently poor methods (FLOWMC $\hat{R} \approx 1.96$, GL-NT ≈ 1.45 on both splits) are poor structurally, not overfit. The one method whose *dev* efficiency is optimistic is the baseline itself (eval/dev = 0.31): the guard-mandated schedule makes S0 the honest, stronger variant, so the "no gradient method beats the baseline" result is if anything conservative.

7.7 The A100 phase (P2c): the two hardest real targets need the recipe and the evidence

The until-converged evaluation on the real 46- and 74-dimensional posteriors returns a sharp — and, for off-the-shelf samplers, negative — result. On **T2** (the condition-10^{14} marginalized posterior, `foundry_marg46`) neither challenger converges cheaply: single-stage preconditioned HMC reaches only $\hat{R}_{\text{mass}} = 3.10$ (ESS 28 at 300 kept draws) and the auto-mass MCLMC challenger — the

Table 6: Development-versus-evaluation generalization of the frozen policies (median over cells; `data/bench_report.json`). Ratios near 1.0 certify no dev→eval overfit.

method	dev ESS/grad	eval ESS/grad	eval/dev	eval $\hat{R}$ (med)
NAUTILUS	3.8×10^{-2}	4.2×10^{-2}	1.10	1.001
S0 baseline	7.9×10^{-3}	2.4×10^{-3}	0.31	1.022
PT-HMC	2.1×10^{-3}	1.5×10^{-3}	0.69	1.004
MCLMC	1.4×10^{-3}	1.5×10^{-3}	1.04	1.010
tempered SMC	1.6×10^{-4}	2.1×10^{-4}	1.29	—
NeuTra	2.4×10^{-4}	2.2×10^{-4}	0.91	1.160
NUTS	3.7×10^{-5}	8.0×10^{-5}	2.15	1.090
GL-NT	7.4×10^{-5}	7.6×10^{-5}	1.03	1.453
FLOWMC	7.4×10^{-5}	7.6×10^{-5}	1.03	1.956

honest "no hand-built mass matrix" contender — is worse at $\hat{R} = 5.9$ (ESS 9). What *does* converge this target is the campaign's own two-stage re-preconditioning recipe (§5.2), which took the same posterior from $\hat{R} = 2.11$ to 1.003. The T2 verdict is therefore that the real ill-conditioned lens posterior *needs* the two-stage recipe, and that vanilla single-stage HMC and auto-mass MCLMC are both insufficient; S0 still mixes better *per draw* than MCLMC ($\hat{R}$ 3.10 vs 5.9), so "hand-built beats auto-mass" holds directionally.

On **T3** (the 74-dimensional bimodal `foundry_v3b74`) *every* direct sampler fails mode recovery: S0-SVI freezes in the steep basin (correct by luck, $\hat{R} = 9.7$, and cannot weight it), the retuned PT-HMC freezes in the *wrong* (low) basin ($\hat{R} \sim 10^{15}$, zero migrations), and NAUTILUS — dominant on the easy and mid targets — *times out after four hours* in the 74-dimensional live-point phase with no output. Only per-basin tempered-SMC evidence yields a trustworthy weight ($\log Z_{\text{steep}} - \log Z_{\text{low}} = +162.2 \pm 1.8$, $w_{\text{steep}} \approx 1$; the same reference used in Table 5), overturning the stored-chain 93.75%-low occupancy as a start artifact with zero inter-basin migrations. The NAUTILUS arc thus closes two-sided: efficiency-dominant where it applies (P2b), but infeasible on *both* hard real regimes — its cond-10^{14} shells collapse and its 74-dimensional bimodal run times out. GL-NT was the drop-first item and is not evaluated on the real targets; combined with its Track-A failures, the pre-registered criteria support a *benchmark* paper, not a *recipe* paper. Both hard-target findings echo the Pillar-1 sampler saga (§6.3) from the sampler side: real lens posteriors are saddle- and multimodal targets that defeat the mock-validated single-metric recipe, and are reached only by re-preconditioning (T2) or by tempered evidence (T3).

8 The CGL Recipe End-to-End, and Euclid Q1

The end-to-end phase combines the settled ingredients — the correlated-noise likelihood (§3) and the two-stage sampler recipe (§5.2) — into a single `build_target` → `map_polish` → `laplace_evidence` → two-stage PHMC → COOLEST pipeline, and applies it both to DESI−165.4754−06.0423 end-to-end and to three Euclid Q1 grade-A lenses (Euclid Collaboration et al., 2025) for an independent real-data test. Two clarifications set expectations. First, Euclid Q1 VIS is native $0.''1$/px imaging, *not* drizzle-resampled, so its instrumental noise is diagonal; the Euclid demonstration therefore exercises the Pillar-2 *sampler* recipe on independent real data with the diagonal likelihood, not the Pillar-1 correlated likelihood (whose real-data test is the HST cross-scale experiment of §6). Second, comparison to the published PyAutoLens catalogue is at the *mass* level (area-equivalent $\theta_{\mathrm{E}}^{\mathrm{eff}}$), where

Table 7: Budget-matched efficiency (median-3-seed ESS/gradient relative to the S0 baseline) on representative targets, with mode recovery on the mixtures. ✓/× under "modes" = both mixture modes recovered within the 0.10 occupancy tolerance. Full matrix: Fig. 6, `data/bench_report.json`. T2/T3 rows summarize the P2c A100 phase (§7.7).

	ESS/grad vs S0			modes	
method	gu-sys3 (hard)	funnel10	illcond46	mix2	mix22
NAUTILUS	180.7	94.4	DQ (shells)	✓	×
S0 baseline	1.00	1.00	**1.00 (owns)**	×	×
PT-HMC	1.87	0.11	blows up	✓	✓
MCLMC	1.08	5.46	0.02 (healthy)	×	×
tempered SMC	0.85	3.38	—	✓	×
NUTS	0.09	0.81	$\hat{R}\,3.8$	×	×
NeuTra	0.56	0.67	$\hat{R}\,5.7$	×	×
FLOWMC	0.41	1.40	$\hat{R}\,2.2$	✓	inverts
GL-NT	0.26	0.28	—	×	×
T2 (real 46d, cond-10^{14})	single-stage HMC $\hat{R}\,3.10$ (ESS 28); auto-mass MCLMC $\hat{R}\,5.9$ (ESS 9) — both fail; only the two-stage recipe converges ($\hat{R}\,2.11 \to 1.003$)				
T3 (real 74d, bimodal)	all direct samplers fail mode recovery (S0 freezes steep, PT freezes low $\hat{R} \sim 10^{15}$, NAUTILUS 4 h timeout); only per-basin SMC weights the modes, $w_{\text{steep}} \approx 1$				

a few-percent convention offset between codes is expected rather than a digit match.

8.1 The recipe runs end-to-end and exports to COOLEST

The recipe is mechanically complete. On DESI–165.4754−06.0423 it runs the full chain on the native `v2d`-relaxed product (Fig. 7): MAP polish, Laplace evidence, two-stage PHMC, source-plane reconstruction, and a standard-format export. Honestly, this native product is the campaign's weakest likelihood (1466 kept pixels), and its posterior is the documented information-limited non-convergence ($\hat{R}_\gamma = 6.0$, $\mathrm{ESS}_{\min} = 30$, the prior-pulled $\gamma \approx 2.23$ of §6); the figure demonstrates the *pipeline*, while the scientifically converged end-state posterior for this system is the binned-low SMC result of §6 (Figs. 4–5). The COOLEST-standard export (Galan et al., 2023) round-trips for all three Euclid targets (`dump_simple` → `load_simple(validate=True)`): the MAP-mode container carries the PEMD+external-shear mass, the four-Sérsic lens light, and the Sérsic+shapelet source, with point estimates preserved to $< 10^{-9}$. Because the 28 source-shapelet amplitudes are *ridge-marginalized* rather than sampled, the marginal-mode $a^\star$ at the MAP is written as the Shapelets point estimate, with the marginalization / $\log\det A$ Occam term and the source-basis mismatch (shapelets versus the published free-form pixelized source) flagged in the COOLEST metadata. Recipe machinery and export are the *working* P3 deliverables.

8.2 Euclid Q1: an honest 1/3-clean characterization

The three grade-A Euclid systems give a mixed, honest outcome (Table 8, Fig. 8), and its interpretation turns on a single diagnostic: *all three MAPs are equally good pixel fits*, with $\chi^2/\mathrm{px} = 0.44$–$0.47$ (verified pixel-exact from the stored posteriors). The θ_E discrepancies below are therefore genuine model degeneracies, *not* failed fits. The well-resolved, round, high-S/N target 102157952 ($\theta_\mathrm{E}^{\mathrm{pub}} = 1.''11$) is a clean success: converged ($\hat{R}_{\theta_\mathrm{E}} = 1.08$, ESS = 449 on only 16 chains), a featureless

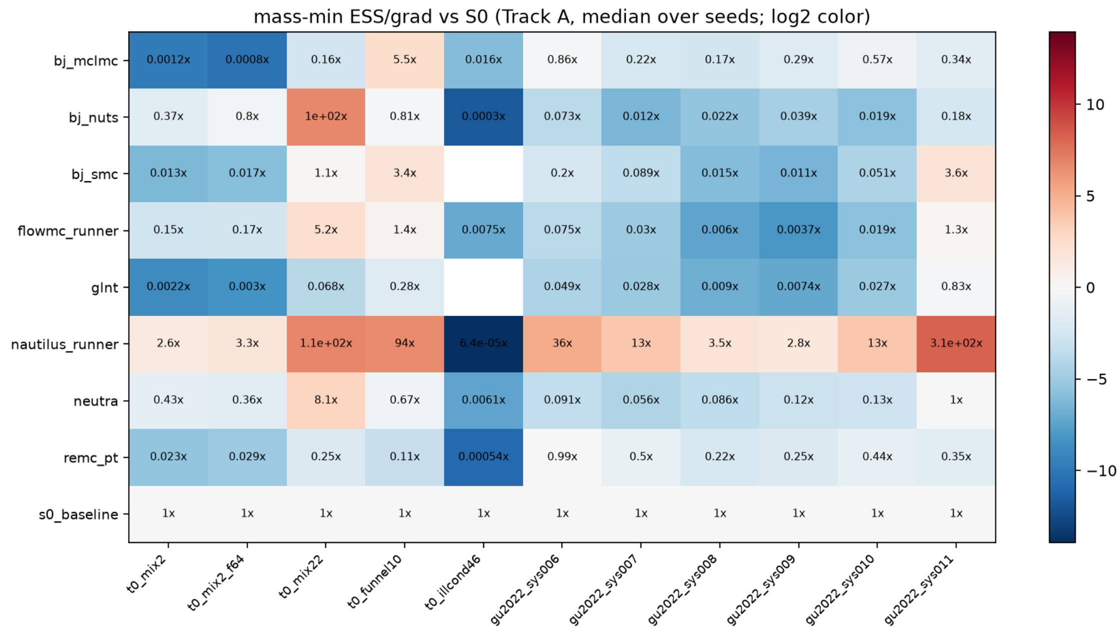

Figure 6: The sampler × target efficiency matrix (Track A, budget-matched, median over three seeds): ESS/gradient relative to the GIGA-Lens baseline, with convergence ($\hat{R}$) shading. NAUTILUS dominates the efficiency columns but is disqualified on cond-10^{14}; the baseline and PT-HMC own reliability; FLOWMC and GL-NT are structurally weak on these geometries. Source: `25_pool_benchmark.py`.

residual with zero $|z| > 3$ pixels, and $\theta_E^{\rm eff} = 1\farcs099$ ($-$**1.4**%, well inside the convention offset). The small, elliptical 102157958 ($\theta_E^{\rm pub} = 0\farcs85$) converges tightly ($\hat{R}_{\theta_E} = 1.04$, ESS $= 2905$) but to $\theta_E^{\rm eff} = 0\farcs581$ ($-$**32**%) with a *better*-than-reference $\chi^2/\text{px} = 0.441$ and a featureless residual: a genuine mass-ellipticity / source-shape degeneracy the data cannot break, not a misfit. The third, 102020061 ($\theta_E^{\rm pub} = 1\farcs24$), does not converge even at 48 chains ($\hat{R}_{\theta_E} = 18.8$): its θ_E posterior is a multimodal ridge spanning $0\farcs79$–$1\farcs28$ (per-chain means spread $0\farcs49$ with within-chain scatter only $\sim 0\farcs03$), a θ_E–ellipticity–centroid degeneracy ($\text{corr}(\theta_E, e_1) = +0.52$, $\text{corr}(\theta_E, c_x) = -0.41$) — so no single θ_E should be quoted. A telling compute paradox: the clean target converged with the *lightest* configuration (16×600), while the two hard targets failed at $3\times$ the budget (48×1400) — more chains cannot fix a degeneracy.

Table 8: Euclid Q1 grade-A fits under the CGL diagonal recipe versus the published PyAutoLens catalogue. All three are equally good pixel fits ($\chi^2/\text{px} \approx 0.45$), so the θ_E offsets are model degeneracies, not misfits. Source: `research/notes/p3-results.md`, `data/euclid/*_fit.json`.

Euclid ID	$\theta_E^{\rm pub}$	$q^{\rm pub}$	$\theta_E^{\rm eff}$ (ours)	offset	χ^2/px	verdict ($\hat{R}_{\theta_E}$)
102157952	$1\farcs11$	0.82	$1\farcs099$	$-$**1.4**%	0.454	**clean** (1.08)
102157958	$0\farcs85$	0.56	$0\farcs581$	-32%	0.441	converged, biased (1.04)
102020061	$1\farcs24$	0.79	$0\farcs79$–$1\farcs28$	ridge	0.472	multimodal (18.8)

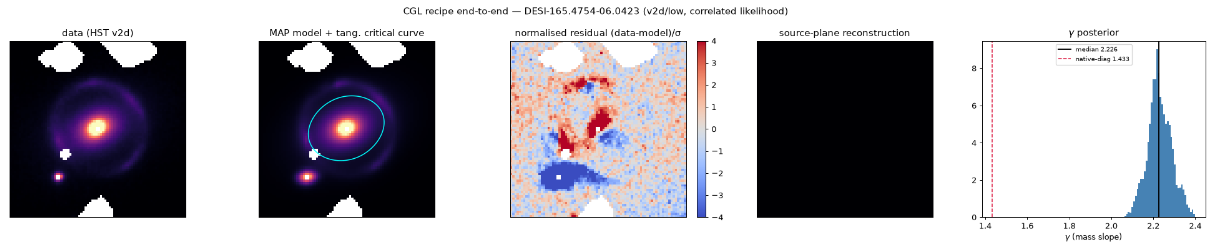

Figure 7: The CGL recipe end-to-end on DESI–165.4754−06.0423 (native `v2d`-relaxed, correlated likelihood): data, MAP model with tangential critical curve, normalized residual, source-plane reconstruction, and the γ posterior. This panel demonstrates that the pipeline (build $\to$ MAP $\to$ Laplace $\to$ two-stage PHMC $\to$ COOLEST) runs mechanically; the γ posterior is the *documented non-convergence* on this information-limited product (1466 kept px, $\hat{R}_\gamma = 6.0$, prior-pulled to ≈ 2.23), and the residual retains structure accordingly. The converged science for this system is Figs. 4–5. Source: `30_recipe_e2e.py`, `data/recipe_e2e/recipe_v2d_low.json`.

8.3 Interpretation and the documented robustness path

Native-resolution single-band VIS modeling is trustworthy for well-resolved, round, high-S/N systems and biased or multimodal for small-θ_E or elliptical ones. Two recurring themes — both pre-registered in the P3 recon and echoing the residual-systematics conclusion of §6 — account for this: PSF / pixel undersampling at $0\overset{\prime\prime}{.}1$/px (a $\theta_E \sim 0\overset{\prime\prime}{.}85$ Einstein ring is only $\sim$1–2 px wide and blends with the lens light; the flat χ^2/px ≈ 0.45 confirms the PSF itself is not oversampled), and source-model choice (our Sérsic+shapelet source selects a different θ_E than the published free-form pixelized source at equal data fit; the source-model systematic of Galan et al. (2024)). The documented robust path — super-sampled ($\times 2$) modeling with a pixelized/free-form source regularized to the PyAutoLens basis — is the route to θ_E robustness and is parked as the P3 follow-on. For this campaign, the recipe and COOLEST export are mechanically validated deliverables, and Euclid θ_E recovery is an honest 1/3-clean characterization of where native-resolution single-band modeling can be trusted.

9 Discussion

The settled results already sharpen the field's picture on both axes. On the likelihood axis, we have shown — with known truth — that ignoring drizzle correlation is not a benign approximation: it biases the recovered slope at $|z| \approx 6$ and destroys coverage, and it does so even at native scale under the real correlation kernel. The correlated likelihood removes that bias and unifies the scales in mocks. On the real system (§6) it proves *necessary but not sufficient*: it dissolves the `v3b` bimodality — per-basin evidence flips by $\sim$191 nats, from favouring the steep basin to favouring the low basin, so the apparent multimodality is a noise-covariance artifact (H1) — but it over-corrects the slope value, driving the binned-low γ to 1.103 ± 0.008, $\sim 17\sigma$ below the diagonal-native anchor, with the correlated slopes bracketing rather than unifying onto it (H2). This is the central, honest result: correcting the noise covariance is a large and real step — it removes the upsampling steep artifact — but a residual scale dependence remains, pointing to source-model and PSF systematics that the shapelet source and stationary-kernel PSF do not yet address, and that the Euclid analysis (§8) independently implicates. The honest standing caveat — the fine posterior is *conservative* by a factor 2.45 because near-singular drizzle covariances lose information at spectral zeros — is itself a useful result: it quantifies the price of correctness, it surfaces on the real fine product as an outright low-basin whitener pathology (§6), and it motivates the λ-sensitivity robustness scan retained as a follow-on.

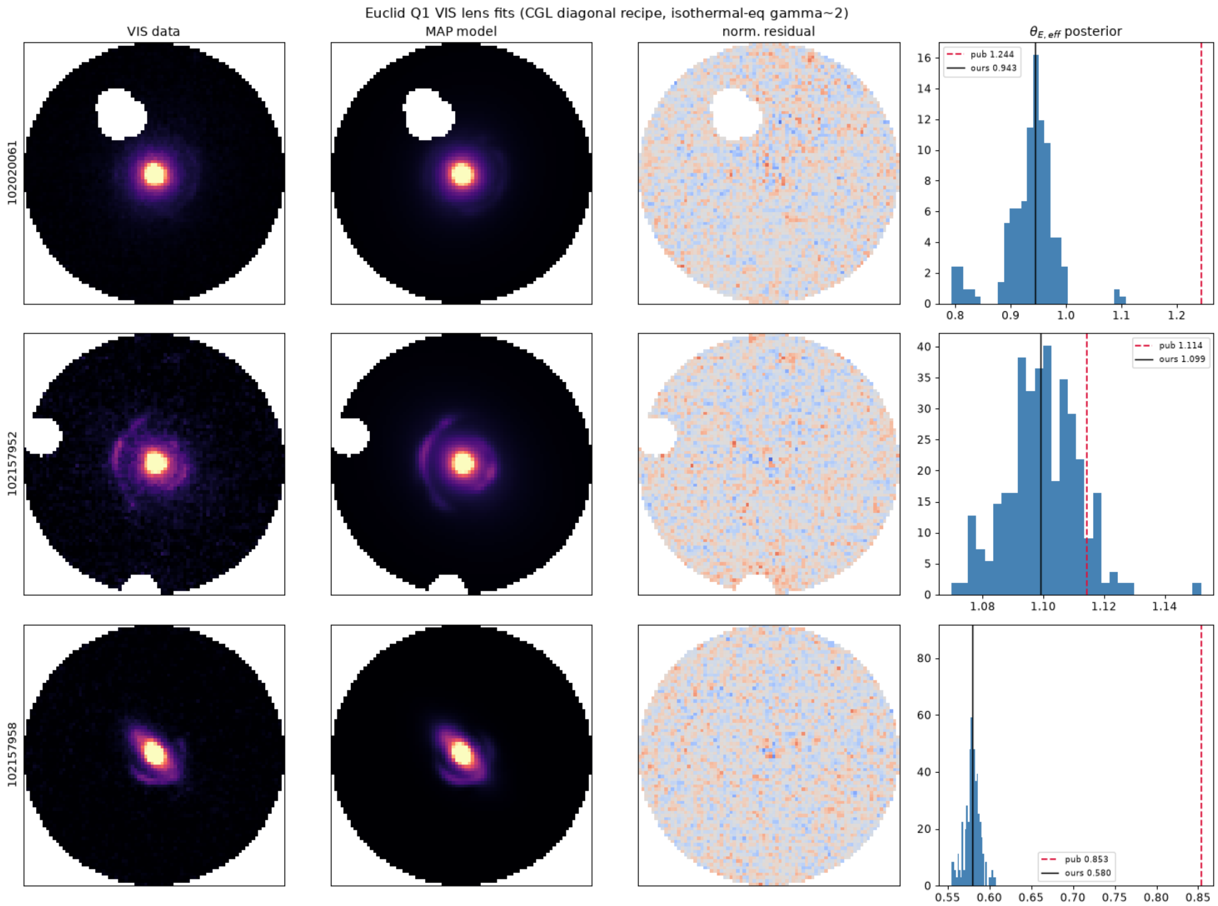

Figure 8: Euclid Q1 VIS fits under the CGL diagonal recipe. Rows: the three grade-A targets; columns: VIS data, MAP model, normalized residual, and the $\theta_{\rm E}^{\rm eff}$ posterior (published value dashed red, ours solid black). 102157952 (middle) is the clean -1.4% success; 102157958 (bottom) is converged but -32% biased (a genuine degeneracy at equal χ^2); 102020061 (top) is the multimodal, non-converged ridge. All three residuals are featureless white noise — the offsets are degeneracies, not misfits. Source: `31_fit_euclid.py`, `research/notes/p3-results.md`.

On the inference axis, the benchmark's two-sided verdict is a caution against single-number sampler comparisons. The most efficient sampler (NAUTILUS) is disqualified precisely on the regime the real marginalized lens posterior occupies; the most reliable until-converged sampler (PT-HMC) is not the most efficient; and no gradient method beats the incumbent baseline at budget parity on the standard mocks. That the ranked flow-assisted recipe (GL-NT) fails its own bars at phoenix budget is a genuine finding, not a null result: it delimits where transport-preconditioned HMC helps and where flow-space trajectory adaptation is the bottleneck. The A100 phase (§7.7) then delivers the sharpest inference lesson: the two hardest *real* lens posteriors defeat every off-the-shelf sampler. The condition-10^{14} marginalized target converges under neither single-stage HMC nor auto-mass MCLMC — only under the campaign's two-stage re-preconditioning recipe — and the bimodal target yields its mode weights to no direct sampler, only to per-basin tempered-SMC evidence. Both findings converge with the Pillar-1 sampler saga (§6.3): real lens posteriors are saddle-shaped and multimodal, and the mock-validated single-metric recipe does not transfer to them — an inference-side result of independent practical value.

Two follow-on pillars are deliberately parked (out of scope, per the campaign's written scope gate) and noted for completeness: a Gaussian-process / pixelated source model (the natural successor to the shapelet source, and a stress test of the correlated likelihood under a flexible source), and higher-order multipole mass structure (Powell et al., 2022; Van de Vyvere et al., 2022), whose $\sim$1%-level effects on γ are of the same order as the correlated-noise systematic quantified here. Both interact with the mass-sheet degeneracy that limits time-delay cosmography (Birrer et al., 2020) and are the subject of a planned P3 follow-on.

10 Reproducibility

Substrate and parity. All experiments run against a vendored, unpatched snapshot of the group production library (`gigalens-sean`, branch `multinode-2025`, ref `58ec9a7`) — bit-identical to the one the Foundry-I reproduction validated — with every campaign modification in a thin `cgl` package whose likelihood and marginalization take array-only APIs (no GIGA-Lens imports), so that any $\Delta\gamma$ is attributable to the likelihood change and not to the substrate. Parity against the validated Foundry-I marginalization core passes all five pre-registered gates at machine precision (Appendix A).

Environment. The pinned stack (identical core to the blessed GIGA-Lens venv: `jax` 0.6.2, `tensorflow-probability` 0.25.0, `numpy` 2.4.6, `scipy` 1.17.1, `optax` 0.2.8, `lenstronomy` 1.14.0, plus `blackjax` 1.3, `flowMC` 0.4.5, `flowjax` 19.0.0, `nautilus-sampler` 1.0.6) is frozen to 147 packages and reproduced with zero resolver drift on both architectures (aarch64 phoenix and Perlmutter x86-A100), the A100 parity re-verified fresh (§A). A single `constraints.txt` pins the sampler dependencies that would otherwise silently upgrade `jax`.

Seven documented stack defects. Reaching the settled state surfaced seven reproducible issues in the `jax` 0.6.2 / TFP 0.25.0 sampler stack (Table 9); they are genuine findings, each with a workaround carried in `cgl`. Two further correctness traps were caught and pinned: an arviz axis-order bug that silently masked stuck chains (a known-$\hat{R}$-2.07 fit reported 1.00 until the `(draw,chain)` axes were corrected), and reversed per-parameter physical labels in the archived Gu et al. (2022) fits (block-contiguous, so aggregate mass sets are unaffected; the zoo uses probed labels).

Provenance and exports. Every headline number in this report cites its source JSON artifact (footnotes and inline `data/*.json`), and every gate is a dated row in `CAMPAIGN.md`, which also carries the retracted intermediate results verbatim (ledger discipline). The parity harness is Appendix A, the compute ledger Appendix B, the zoo specification Appendix C, and the frozen sampler configurations Appendix D. The end-state products will be exported to the code-independent COOLEST standard (Galan et al., 2023) in P3 (§8).

A Pre-Registered Verification Thresholds and Parity

Table 10 reproduces the pre-registered thresholds verbatim from `README.md` §Verification (frozen at P0, before any science ran). Table 11 gives the measured parity-harness values.

Table 9: The seven stack defects found and fixed during the campaign (detail in `CAMPAIGN.md`). All are in the `jax` 0.6.2 / `jaxlib` 0.6.2 / TFP 0.25.0 stack.

defect	workaround
#1 `chex` 0.1.92 silently upgrades `jax` 0.6.2→0.10.2, breaking TFP	pin via `constraints.txt`; all installs pass `-c`
#2 XLA `priority-fusion` livelocks fusing f64 `random.normal` with a reduction (MCLMC momentum refresh)	`XLA_FLAGS=--xla_disable_hlo_passes=priority-fusion`, set pre-import
#3 `jaxlib` 0.6.2 triton GEMM aborts on tiny f32 dots (dim-2 SVI) on L4	`--xla_gpu_enable_triton_gemm=false` (f64 immune)
#4 priority-fusion livelock recurs in f32 (MCLMC tuner / NSF pipelines)	extend the flag to those adapters' processes
#5 SMC particle-with-gradient memory ceiling ∼21 MB/particle (1200 particles OOM a 24 GiB L4)	cap particle count per device; frozen at 384/L4
#6 NAUTILUS jitted `log_like` `TracerArrayConversion` on new batch sizes	batch padding + bijector-template probe
#7 correlated-marg batched gradient (28 per-column convs under `vmap`) livelocks XLA compile	collapse to one depthwise/grouped conv; logp bit-identical, compiles in 13.8 s

B Compute Ledger

All Perlmutter jobs charge account `deepsrch_g` (P1c/P2c ran on `cosmo_g` during a scheduling-latency exception; see `CAMPAIGN.md`); each ledger row is appended before results are read. Committed ≤ 90 A100-h of the repo-wide allocation, hard stop 100; the campaign used ∼34 A100-h in total (P0 1.2 + P1c smoke ≤ 0.5 + P1c ∼15 + P2c ∼17), well within budget, with Euclid (P3) run budget-free on phoenix L4.

C Posterior-Zoo Target Specifications

The 19 frozen targets (`data/zoo_validation.json`) and their five acceptance gates: (i) T1 zoo `log_prob` bit-identical to direct construction (max rel 0.0); (ii) T2 `log_prob` reproduces the stored value to $\leq 10^{-8}$ ($\log p = -45840.984005998456$); (iii) T3 mass bijector reproduction 0.0 with basins measured 45 low / 3 steep ($\bar{\gamma}^{\text{low}} = 1.294$, occupancy 0.9375/0.0625, zero migrations); (iv) T0 analytics recovered (logZ / mode occupancy, $\hat{R} < 1.001$); (v) prior+like=prob consistency, worst relative 4.9×10^{-8} (f32).

D Frozen Sampler Configurations

The frozen policies (`data/policies_frozen.json`, `2026-07-06T22:08Z`; ≤ 8 configs per method tuned on the dev split only) are summarized in Table 14; each carries a content-addressed `sha_config` and a rationale in the JSON. T0/T1 grad budgets are $2 \times 10^5/1.5 \times 10^6$; Track B scales these $\times 4$ with configuration pinned.

Table 10: Pre-registered verification thresholds (verbatim, `README.md` §Verification). Goalposts fixed before any science run; changes require a dated gate exception in `CAMPAIGN.md`.

gate	threshold	status
A forward image	$\max|\Delta|/\max|\text{img}| \le 10^{-12}$ (f64)	PASS
B log-posterior	$|\Delta \log p| \le 10^{-8}$	PASS
C gradient	relative $L_2 \le 10^{-8}$	PASS
D diagonal limit	corr. with $C = \text{diag}$ equals diagonal stack $\le 10^{-10}$	PASS
E Occam term	$-\frac{1}{2}\log\det A$ vs numpy `slogdet` $\le 10^{-10}$	PASS
whitener e_{op}	≤ 0.02	PASS
whitener MC whiteness	$\text{Var}(u) \in 1 \pm 0.02$; $|\text{off-diag}|_{\le 3} < 0.01$	PASS
whitener $|\Delta \log L|$	< 0.1 nat vs dense-C, $80^2/130^2$, 20 draws	PASS
kernel fit	$\max|\rho_{\text{fit}} - \rho_{\text{meas}}| \le 0.05$, model-subtracted	PASS
mock render	$< 0.05\sigma$ on every kept pixel	PASS
E1a artifact	median $|z(\gamma)| > 2$ OR $\text{cov}_{68} < 40\%$	PASS
E1b recovery	$|\bar{z}(\gamma)| < 0.5$/scale; cross-scale $\ge 6/8$; width $\in [0.7, 1.5]$	PASS*
E1c SBC (64 mocks)	rank $p > 0.01$ each of 6; $\text{cov}_{68} \in [55, 80]\%$	PASS*
E2 H1 bimodality	steep mass $< 10\%$ OR corrected $\Delta \log \ell \le 0$ (both branches)	**PASS** (artifact; $\Delta\log Z{=}{-29}$)
E2 H2 cross-scale	$|\Delta\gamma| < 2\sigma_{\text{comb}}$; anchor 1.433 [1.400, 1.469]	FAIL (over-corr.; 1.103, $\sim 17\sigma$)
E2 H3 honesty	$\sigma_\gamma(\text{fine}) \ge \sigma_\gamma(\text{native})/1.5$	**PASS** ($0.117 \ge 0.023$)
E3 robustness	γ-mean shift $< 0.5\sigma$ across kernel variants	follow-on (not run)
P2 win criterion	ESS/grad $\ge 2\times$ AND $\hat{R} < 1.01$ AND modes ± 10 pts	applied
budget	Perlmutter commit 90 A100-h, hard stop 100, `deepsrch_g`	on track

Note. *E1b width ratio and E1c full-set γ rank are met with the documented caveats of §5 (conservative-not- biased width; healthy-fit restriction); the depth-controlled amendments are recorded in `CAMPAIGN.md`.

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Table 11: Parity-harness measurements versus the validated Foundry-I `_hmc_lib_marg` core, at `map_marg_pd.npz` plus three seeded perturbations and the Gu et al. (2022) `system_000` truth (`data/parity_report.json`).

gate	threshold	achieved	note
A forward image	10^{-12}	6.2×10^{-17}	f64, harness-rebuilt m_{det}
B log-posterior	10^{-8}	0.0	worst of 4 points
C gradient rel-L_2	10^{-8}	7.1×10^{-17}	grad-norm 400.5806 at z_{ref}
D diagonal limit	10^{-10}	0.0	conv-whitener = diagonal exactly
E Occam term	10^{-10}	0.0	$\log\det A = 323.229$, $\text{cond}(A) = 1.4 \times 10^4$
(info) stored MAP logp	10^{-5}	6.5×10^{-11}	pip-era A16 vs vendored L4 drift
(info) gu-2022 forward	10^{-5}	1.5×10^{-6}	f32 stored render

Table 12: Compute to the settled state. Phoenix is 8×A16 15 GB +2×L4 23 GB (budget-free, weekly-rolled); Perlmutter is 4×A100 per node.

phase	platform	cost	note
P0 staging + A100 parity	Perlmutter	1.2 A100-h	parity A–E PASS both arch
P1c smoke (pre-flight)	Perlmutter	$\leq$ 0.5 A100-h	shared QOS
P1a–P1b (kernels, mocks, E1, diagnosis)	phoenix	∼165 GPU-h	incl. ∼95 diagnosis
P2a–P2b (zoo, benchmark)	phoenix	202.4 GPU-h	370 cells
P1c cross-scale	Perlmutter	∼15 A100-h	SMC money product; $\leq$ 30 committed
P2c sampler matrix	Perlmutter	∼16–18 A100-h	seed-0 T2/T3; $\leq$ 45 committed
P3 recipe + Euclid	phoenix	∼0 A100-h	Euclid on phoenix L4 (diagonal)

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Table 13: The posterior zoo. dtype is the frozen evaluation precision; identity tolerance is the adapter-fidelity assertion at three reference points.

target	tier	dtype	description
`t0_mix2 / _f64`	T0	f32/f64	two-component Gaussian mixture (0.8/0.2), analytic Z + modes
`t0_mix22`	T0	f32	well-separated mixture (the hard mode-recovery target)
`t0_funnel10`	T0	f32	10-d Neal funnel (varying curvature)
`t0_illcond46`	T0	f64	46-d ill-conditioned Gaussian, cond $\sim 10^{14}$ (metric stress)
`gu2022_sys000--011`	T1	f32	twelve Gu et al. (2022) mock lens posteriors
`foundry_marg46`	T2	f64	the 46-d marginalized real posterior of DESI–165.4754−06.0423 (cond 10^{14})
`foundry_v3b74`	T3	f32	the 74-d genuinely bimodal binned real posterior
(Euclid Q1 $\times$3)	T4	—	fit end-to-end in §8 (native diagonal, not benchmarked as zoo cells)

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Table 14: Frozen T1 sampler configurations (representative fields; `data/policies_frozen.json`).

method	frozen configuration (key fields)
S0 baseline	MAP→SVI→ChEES-HMC; 50 chains ×(250 burn +750 keep); guard-floored SVI
BlackJAX NUTS	SVI start, target-accept 0.8, tree-depth cap 2^6 (bounds grad budget), diagonal mass
BlackJAX MCLMC	unadjusted, tuner preconditioning, SVI-diagonal mass fallback for the cond-10^{14} tuner collapse
BlackJAX SMC	adaptive tempered, target-ESS 0.7, 3 MCMC steps, 5 leapfrog, 1000 particles
flowMC	prior start, RQSpline (4 layers, 8 bins, 64×64), scalar MALA step 0.6, 32 chains
nautilus	1000 live, 4 networks, $f_{\text{live}} = 0.01$, cap = 2× grad budget (gradient-free)
PT-HMC (remc_pt)	6 replicas, $\beta_{\min} = 0.2$ (T1), 8 leapfrog, SVI mass; T3 ladder to be re-tuned (P2c)
NeuTra	SVI-flow pullback (6 layers, 8 knots) then ChEES-HMC in flow space
GL-NT	MAP→SVI→SMC anneal → NSF fit → flow-space ChEES-HMC (the recipe under test)